

Quantitative Risk Assessment Using Indicators and Warning (I&W) Analysis

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June 15, 2026

Introduction

This report applies the Indicators and Warning (I&W) methodology to quantitatively assess three critical risks associated with TD Bank's failed acquisition of First Horizon Bank. As identified in the previous Risk Assessment and Risk Register, the acquisition was significantly affected by economic uncertainty, prolonged regulatory scrutiny, and strategic execution challenges. The three risks selected for quantitative assessment are Regulatory Approval Failure, Market Share Loss, and Reputational Damage.

The I&W methodology was selected because it focuses on identifying leading indicators that provide early warning signals before risks fully materialize. Each risk was evaluated using measurable indicators, threshold levels, trend analysis, and quantitative forecasting techniques. In addition, Monte Carlo simulation was considered to estimate the probability of adverse outcomes under varying scenarios.

Risk 1: Regulatory Approval Failure

The most significant risk associated with the First Horizon acquisition was regulatory approval failure. The leading indicator selected for this risk was Regulatory Review Duration, measured as the number of months required for regulatory approval of major banking acquisitions.

TD announced the acquisition of First Horizon on February 28, 2022. The transaction was initially expected to close within approximately twelve months. However, on February 9, 2023, TD and First Horizon announced an extension of the merger agreement due to ongoing regulatory review. The acquisition was ultimately terminated on May 4, 2023 because the parties could not obtain regulatory approval within an acceptable timeframe.

Indicator and Thresholds

Indicator	Green	Yellow	Red
Regulatory Review Duration	Less than 12 months	12–15 months	More than 15 months

The actual review period exceeded fourteen months and approached the Red warning threshold. This indicated increasing regulatory uncertainty and a growing probability of deal failure.

To quantitatively assess this risk, a forecasting model would be developed using historical merger approval timelines within the banking industry. A Monte Carlo simulation could then be performed using approval duration as the primary variable. Simulation results would estimate the probability that future regulatory reviews exceed critical thresholds. Based on observed delays and eventual deal termination, this risk was assessed as having a high probability and high impact.

Risk 2: Market Share Loss

The second risk identified was market share loss resulting from the failed acquisition. One of the strategic objectives of the First Horizon transaction was to expand TD Bank's presence throughout the southeastern United States. Failure to complete the acquisition limited TD's planned growth opportunities and may have weakened its competitive position within key markets.

The leading indicator selected for this risk was Customer Churn Rate. Customer churn is widely recognized as a leading indicator because increasing customer attrition often precedes declines in market share and revenue performance.

Indicator and Thresholds

Indicator	Green	Yellow	Red
Customer Churn Rate	Less than 3%	3%–5%	More than 5%

To assess this risk quantitatively, customer growth trends, deposit growth rates, and market share statistics from TD annual reports and FDIC industry reports would be analyzed. A regression model could be applied to forecast future customer retention trends under different competitive scenarios.

Monte Carlo simulation would then be used to estimate the probability of customer churn exceeding the defined threshold levels. Simulation results would provide management with an estimate of potential customer losses and their impact on future growth objectives. Given the strategic importance of the acquisition, this risk was assessed as having a moderate-to-high probability and impact.

Risk 3: Reputational Damage

The third risk examined was reputational damage resulting from the failed acquisition and the negative publicity surrounding prolonged regulatory review. Reputational risk is particularly important in the banking industry because customer trust, investor confidence, and regulatory relationships directly influence long-term performance.

The leading indicator selected for this risk was Negative Media Coverage. Increased negative media attention often serves as an early warning signal of reputational deterioration before financial consequences become visible.

Indicator and Thresholds

Indicator	Green	Yellow	Red
Negative Media Coverage	Less than 5 major negative reports	5–10 reports	More than 10 reports

Following the announcement delay and eventual termination of the transaction, major media outlets including Reuters, Bloomberg, and S&P Global published extensive coverage regarding regulatory concerns and strategic uncertainty surrounding TD Bank.

To quantitatively assess this risk, historical media coverage and public sentiment trends would be monitored over time. Forecasting techniques could be used to estimate future negative media exposure, while Monte Carlo simulation would estimate the probability of exceeding established warning thresholds. Given the visibility of the failed transaction and its impact on investor confidence, this risk was assessed as having a high impact and moderate-to-high probability.

Conclusion

The Indicators and Warning (I&W) analysis identified Regulatory Approval Failure as the highest-priority risk due to its direct contribution to the termination of the First Horizon acquisition. Market Share Loss and Reputational Damage were also identified as significant risks because of their potential long-term effects on TD Bank's growth strategy and stakeholder confidence.

By establishing measurable leading indicators, defining warning thresholds, and incorporating forecasting and Monte Carlo simulation techniques, TD Bank can improve its ability to detect emerging threats and support more informed strategic decision-making. The analysis demonstrates how quantitative risk assessment can complement qualitative risk management practices and strengthen organizational resilience during periods of economic and regulatory turbulence.

References

1. Federal Reserve. (2023). Bank merger applications and regulatory review process.
<https://www.federalreserve.gov>
2. First Horizon Corporation. (2023). Investor relations and merger updates.
<https://ir.firsthorizon.com>
3. TD Bank Group. (2023). TD and First Horizon extend outside date of proposed transaction.
<https://stories.td.com/ca/en/news/2023-02-09-td-and-first-horizon-extend-outside-date-of-proposed-transac>
4. TD Bank Group. (2023). TD Bank and First Horizon mutually agree to terminate merger agreement.
<https://stories.td.com/ca/en/news/2023-05-04-td-bank-and-first-horizon-mutually-agree-to-terminate-merger>
5. TD Bank Group. (2023). Annual Report 2023.
<https://www.td.com/ca/en/about-td/for-investors>
6. Reuters. (2024). TD CEO's compensation reduced following regulatory issues and failed First Horizon acquisition.